

Calculus II Practice Final — Worked Answer Key

A balanced twenty-five-question Calculus II final with study map and complete worked key.

Revision: 2026-07-23 **Canonical page:** <https://bettergrades.net/subjects/math/calculus/practice-exams/calculus-2-final/>

1. Evaluate $\int x e^x dx$.

Answer: $e^x(x - 1) + C$

- Use integration by parts.
- The verified result is $e^x(x - 1) + C$.

2. Evaluate $\int \sin^3 x \cos x dx$.

Answer: $\sin^4 x / 4 + C$

- Use $u = \sin x$.
- The verified result is $\sin^4 x / 4 + C$.

3. Evaluate $\int 1/(x^2 - 1) dx$.

Answer: $\frac{1}{2} \ln \left| \frac{x-1}{x+1} \right| + C$

- Use partial fractions.
- The verified result is $\frac{1}{2} \ln \left| \frac{x-1}{x+1} \right| + C$.

4. Determine convergence of $\int_1^\infty x^{-2} dx$.

Answer: *convergesto1*

- Evaluate the defining limit.
- The verified result is *convergesto1*.

5. Find area between $y=x$ and $y=x$ squared on $[0,1]$.

Answer: $1/6$

- Integrate top minus bottom.
- The verified result is $1/6$.

6. Rotate $y=x$ on $[0,2]$ about x-axis. Find volume.

Answer: $8\pi/3$

- Use disks: $x^2 dx$.
- The verified result is $8\pi/3$.

7. A force $F(x)=4x$ acts from $x=0$ to 3. Find work.

Answer: 18

- Integrate force over displacement.
- The verified result is 18.

8. Find $\lim n/(n+1)$.

Answer: 1

- a. Divide by n .
- b. The verified result is 1.

9. Sum $\sum_{n=0}^{\infty} 3(1/2)^n$.

Answer: 6

- a. Use $a/(1-r)$.
- b. The verified result is 6.

10. Test $\sum n/(n+1)$ for convergence.

Answer: *diverges*

- a. Terms do not approach zero.
- b. The verified result is *diverges*.

11. Test $\sum 1/n^{3/2}$.

Answer: *converges*

- a. It is a p -series with $p > 1$.
- b. The verified result is *converges*.

12. Test $\sum 1/(n^2 + 4)$.

Answer: *converges*

- a. Compare with $1/n$ squared.
- b. The verified result is *converges*.

13. Test $\sum (3n+1)/(n^2+2)$.

Answer: *diverges*

- a. Limit compare with $1/n$.
- b. The verified result is *diverges*.

14. Test $\sum n!/4^n$.

Answer: *diverges*

- a. The ratio $(n+1)/4$ eventually exceeds 1.
- b. The verified result is *diverges*.

15. Test $\sum (2n/(3n+1))^n$.

Answer: *converges*

- a. Root-test limit is $2/3$.
- b. The verified result is *converges*.

16. Test $\sum(-1)^{n-1}/n$.

Answer: *converges conditionally*

- a. AST applies; harmonic absolute series diverges.
- b. The verified result is *converges conditionally*.

17. Classify $\sum(-1)^n/n^2$.

Answer: *converges absolutely*

- a. The absolute series is a p-series with $p=2$.
- b. The verified result is *converges absolutely*.

18. Find R for $\sum x^n/5^n$.

Answer: $R = 5$

- a. It is geometric with ratio $x/5$.
- b. The verified result is $R = 5$.

19. Find interval for $\sum(x-1)^n/n$.

Answer: $[0, 2)$

- a. $R=1$, then test both endpoints.
- b. The verified result is $[0, 2)$.

20. Write the Maclaurin series for $\cos x$.

Answer: $\sum(-1)^n x^{2n}/(2n)!$

- a. Use the derivative cycle.
- b. The verified result is $\sum(-1)^n x^{2n}/(2n)!$.

21. Find T_3 for e to the power x at 0.

Answer: $1 + x + x^2/2 + x^3/6$

- a. Use derivatives of e to the power x .
- b. The verified result is $1 + x + x^2/2 + x^3/6$.

22. Bound alternating-series error after four terms of $\sum(-1)^{n-1}/n$.

Answer: $\leq 1/5$

- a. Use the first omitted term.
- b. The verified result is $\leq 1/5$.

23. For $x=t$ squared, $y=t$ cubed find dy/dx .

Answer: $3t/2$

- a. Divide dy/dt by dx/dt .
- b. The verified result is $3t/2$.

24. Convert $r=2\cos$ to Cartesian form.

Answer: $(x - 1)^2 + y^2 = 1$

a. Use $r^2 = 2r \cos$.

b. The verified result is $(x - 1)^2 + y^2 = 1$.

25. Why must power-series endpoints be tested separately?

Answer: *The ratio/root test is inconclusive when $|x - a| = R$.*

a. At the boundary the limiting ratio equals 1.

b. The verified result is *The ratio/root test is inconclusive when $|x - a| = R$.*